



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

General Chemistry Final Exam

Course Name: General Chemistry (CH103) Dept.: Chemistry

Exam Duration: 2h

Exam Paper Setter: XZX, BT, ZYZ, ZWQ, LHZ, LL, JG, JW ..

Part No.	A	B	C
Score	50	16	34

This exam paper contains 3 parts (Part A: 25 questions, Part B: 4 questions and Part C: 4 questions) and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Part A: Multiple Choice - Select the one that is best in each case. (50 Pts)

Number of questions: 25

Points per question: 2

Part B: Short Answer -- Write legibly and show all work for all steps in the problem. (16 Pts)

Number of questions: 4

Points per question: 4

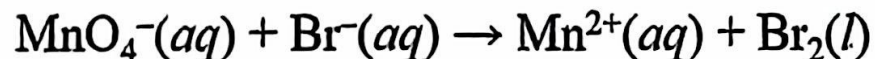
Part C: Long Answer -- Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (34 Pts) Number of questions: 4 (8 + 8 + 10 + 8 Points)

Part A: Multiple Choice – Select the one that is best in each case. (40 Pts)

Number of questions: 20

Points per question: 2

A1. How many water molecules are there in the equation when the following equation is balanced (for the reaction balanced with the smallest whole-number coefficients)?



- A) Four on the reactant side **B) Eight on the product side**
C) One on the reactant side D) Seven on the product side
E) Eight on the reactant side

2. Arrange the following substances in order from weakest to strongest acid:



- A) $\text{HIO}_2 < \text{HClO}_3 < \text{HBrO}_3 < \text{HIO}_3 < \text{HClO}_4$ B) $\text{HIO}_3 < \text{HIO}_2 < \text{HClO}_4 < \text{HBrO}_3 < \text{HClO}_3$
C) $\text{HClO}_4 < \text{HClO}_3 < \text{HBrO}_3 < \text{HIO}_3 < \text{HIO}_2$ D) $\text{HBrO}_3 < \text{HIO}_3 < \text{HClO}_3 < \text{HIO}_2 < \text{HClO}_4$
E) $\text{HIO}_2 < \text{HIO}_3 < \text{HBrO}_3 < \text{HClO}_3 < \text{HClO}_4$

A3. What information about water is needed to calculate the enthalpy change for converting 1 mol $\text{H}_2\text{O}(g)$ at 100 °C to $\text{H}_2\text{O}(l)$ at 80 °C?

- A) heat of fusion
B) heat of vaporization
C) heat of vaporization and specific heat of $\text{H}_2\text{O}(g)$
D) heat of vaporization and specific heat of $\text{H}_2\text{O}(l)$
E) heat of fusion and specific heat of $\text{H}_2\text{O}(l)$

Part D. Short Answer – write legibly in each of the following problems. (30 Pts)

Number of questions: 6 Points per question: 5

B1. Based on the type or types of intermolecular forces, predict and explain the substance in each pair that has the higher boiling point:

(a) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$

Part A: Multiple Choice - Select the one that is best in each case. (40 Pts)

Number of questions: 20

Points per question: 2

A1. What types of intermolecular forces exist between NH_3 and H_2S ?

- A) dispersion forces and hydrogen bonds
- B) dispersion forces, dipole-dipole forces, and hydrogen bonds
- C) dispersion forces, hydrogen bonds, and ion-dipole forces
- D) dispersion forces E) dispersion forces and dipole-dipole forces

A2. Which statement is true about liquids but not true about solids?

- A) They flow and are highly ordered.
- B) They are highly ordered and not compressible.
- C) They flow and are not compressible.
- D) They assume both the volume and the shape of their containers.
- E) They flow and are compressible.

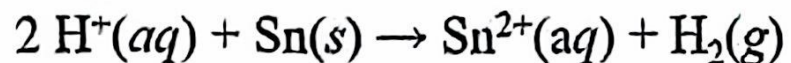
A3. Calculate the enthalpy change (in kJ) associated with the conversion of 18.0 grams of ice at -4.00°C to water at 40.00°C . The specific heats of ice and water are $2.09\text{ J/g}\cdot\text{K}$, and $4.18\text{ J/g}\cdot\text{K}$, respectively. For H_2O , $\Delta H_{\text{fus}} = 6.01\text{ kJ/mol}$.

- A) 6.01 B) 9.17 C) 9170 D) 3.16 E) 3160

A4. A solution is prepared by dissolving 15.0 g of NH_3 in 250.0 g of water. The density of the resulting solution is 0.974 g/mL . The molarity of NH_3 in the solution is _____ M.

- A) 3.53×10^{-3} B) 60.0 C) 8.82×10^{-1} D) 3.24 E) 3.53

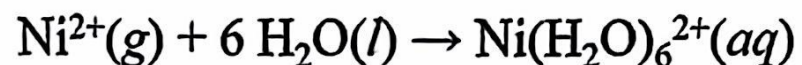
119. Consider an electrochemical cell based on the reaction:



Which of the following actions would not change the measured cell potential?

- a) lowering the pH in the cathode compartment
- b) addition of more tin metal to the anode compartment**
- c) increasing the tin (II) ion concentration in the anode compartment
- d) increasing the pressure of hydrogen gas in the cathode compartment
- e) Any of the above will change the measured cell potential.

120. In the following reaction, Ni^{2+} is acting as a(n) _____.



- A) solvent B) precipitating agent **C) Lewis acid** D) oxidizing agent E) ligand

C3. (a) Write the chemical equation that defines the normal boiling point of liquid carbon tetrachloride, $\text{CCl}_4(l)$.

(b) What is the value of ΔG° for the equilibrium in question (a)?

(c) Use data below to estimate the normal boiling point of $\text{CCl}_4(l)$.

Substance	ΔH_f° (kJ/mol)	S° (J mol ⁻¹ K ⁻¹)
$\text{CCl}_4(g)$	-106.7	309.4
$\text{CCl}_4(l)$	-139.3	214.4

Answer:

(a) $\text{CCl}_4(l) \rightleftharpoons \text{CCl}_4(g) \quad P = 1 \text{ atm}$

(b) At equilibrium, $\Delta G^\circ = 0$

(c) $\Delta G^\circ = \Delta H^\circ - T_b \Delta S^\circ = 0$

$$\Delta H^\circ = (1 \text{ mol})(-106.7 \text{ kJ/mol}) - (1 \text{ mol})(-139.3 \text{ kJ/mol}) = +32.6 \text{ kJ}$$

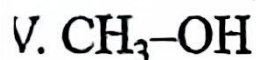
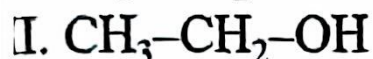
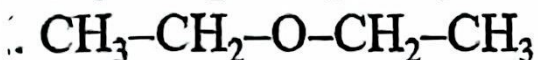
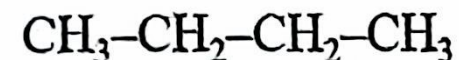
$$\Delta S^\circ = (1 \text{ mol})(309.4 \text{ J/mol-K}) - (1 \text{ mol})(214.4 \text{ J/mol-K}) = +95.0 \text{ J/K}$$

$$T_b = \frac{\Delta H^\circ}{\Delta S^\circ} = \left(\frac{32.6 \text{ kJ}}{95.0 \text{ J/K}} \right) \left(\frac{1000 \text{ J}}{1 \text{ kJ}} \right) = 343 \text{ K} = 70^\circ \text{C}$$

4. The magnitude of K_w at 25°C indicates that _____.

- A) water autoionizes very quickly B) water autoionizes only to a very small extent
C) the autoionization of water is exothermic
D) water autoionizes very slowly E) None of the above is true.

5. Rank the following compounds according to increasing solubility in water.



- A) I < III < IV < II B) I < II < IV < III C) III < IV < II < I

- D) I < II < III < IV E) None of these is correct.

6. The reaction $A \rightarrow B$ is first order reaction in $[A]$. Consider the data below.

Time (s)	$[A]$ (M)
0.0	2.6
10.0	0.65
20.0	0.16

The half-life of this reaction is _____ s.

- A) 0.14 B) 0.97 C) 3.0 D) 5.0 E) 7.1

A7. The standard cell potential (E°_{cell}) for the voltaic cell based on the reaction below is _____ V.



Half-reaction	E° (V)
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.440
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{s})$	+0.771
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.154

- A) 0.46 **B) 0.617** C) 0.925 D) -0.46 E) 1.21

A8. Which compound listed below has the **greatest** molar solubility (mol/L) in water?

- A) CdCO_3 ($K_{\text{sp}} = 1.8 \times 10^{-14}$) B) $\text{Cd}(\text{OH})_2$ ($K_{\text{sp}} = 2.5 \times 10^{-14}$)
C) AgI ($K_{\text{sp}} = 8.3 \times 10^{-17}$) **D) CaF_2 ($K_{\text{sp}} = 3.9 \times 10^{-11}$)** E) ZnCO_3 ($K_{\text{sp}} = 1.0 \times 10^{-10}$)

A9. In which one of the following solutions is silver bromide the most soluble?

- A) 0.200 M HCl B) 0.750 M LiNO_3 C) pure H_2O
D) 0.185 M KCl **E) 0.0150 M NH_3**

A10. For the hypothetical reactions I and II, $K_1 = 10^2$ and $K_2 = 10^{-4}$.

- I. $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2 \text{AB}(\text{g})$
II. $2 \text{A}_2(\text{g}) + \text{C}_2(\text{g}) \rightleftharpoons 2 \text{A}_2\text{C}(\text{g})$
III. $\text{A}_2\text{C}(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2 \text{AB}(\text{g}) + (1/2) \text{C}_2(\text{g})$

What is the value of K for reaction III?

- A) 10^{-2} **B) 10^4** C) 10^6 D) 10^2 E) 10^{-4}

A11. Which one of the following pairs **cannot** be mixed together to form a buffer solution?

- A) NH_3 , NH_4Cl B) $\text{NaC}_2\text{H}_3\text{O}_2$, HCl ($\text{C}_2\text{H}_3\text{O}_2^-$ = acetate)
C) **RbOH , HBr** D) KOH , HF E) H_3PO_4 , KH_2PO_4

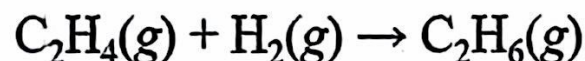
A12. For a reaction in a voltaic cell, both ΔH° and ΔS° are positive. Which of the following statements is true?

- A) E°_{cell} **will increase with an increase in temperature.**
B) E°_{cell} will not change when the temperature increases.
C) E°_{cell} will decrease with an increase in temperature.
D) $\Delta G^\circ > 0$ for all temperatures.
E) None of the above statements is true.

A13. A solution is prepared by dissolving 0.60 g of nicotine (a nonelectrolyte) in water to make 12 mL of solution. The osmotic pressure of the solution is 7.55 atm at 25°C . The molar mass of nicotine is ____ g/mol.

- A) 28 B) 43 C) 85 **D) 1.6×10^2** E) 270

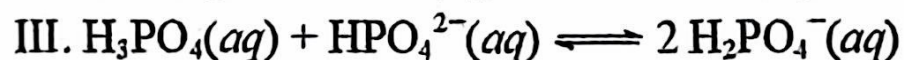
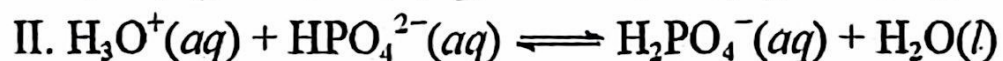
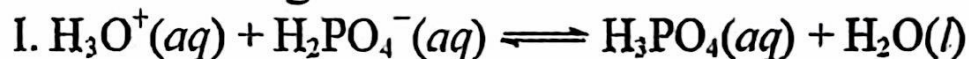
A14. The value of ΔG° at 25°C for the following reaction:



is _____ kJ. At 298 K, ΔH° for this reaction is -137.5 kJ , and ΔS° is -120.5 J/K .

- A) -3.605×10^4 B) -173.4 C) 3.577×10^4 **D) -101.6** E) 101.6

Q15. The dihydrogenphosphate ion, H_2PO_4^- is amphiprotic. In which of the following reactions is/are this ion serving as a **base**?



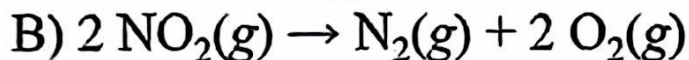
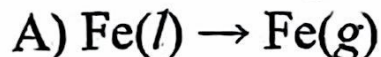
- A) I only B) I and II **C) I and III** D) II and III E) All are correct.

Q16. The following equilibrium is exothermic. How could the yield of diiodine pentoxide be increased?

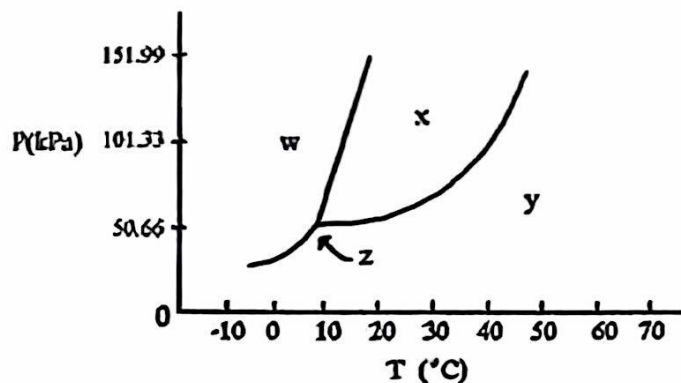
$$\text{I}_2(\text{g}) + 5\text{CO}_2(\text{g}) \rightleftharpoons 5\text{CO}(\text{g}) + \text{I}_2\text{O}_5(\text{g})$$

- A) by decreasing the pressure B) by increasing the pressure
C) by decreasing the volume of the reaction vessel
D) by decreasing the temperature E) by increasing the temperature

Q17. ΔS is **negative** for the process _____.



A18. According to the phase diagram shown below, what is the normal melting point (°C)?



- A) 15 B) 25 C) 35 D) 45 E) 55

A19. The second law of thermodynamics can be given as _____.

- A) $\Delta E = q + w$
 B) $\Delta H^\circ_{\text{rxn}} = \sum n\Delta H^\circ_f(\text{products}) - \sum m\Delta H^\circ_f(\text{reactants})$
 C) for any spontaneous process, the entropy of the universe increases
 D) the entropy of a pure crystalline substance at absolute zero is zero
 E) $\Delta S = q_{\text{rev}}/T$ at constant temperature

20. Which of the following statements concerning a voltaic cell is/are correct?

- I. Oxidation occurs at the anode.
- II. A spontaneous reaction generates an electric current in a voltaic cell.
- III. Anions in the salt bridge flow toward the anode during discharge.

- A) I only B) II only C) III only D) I and II E) All are correct.

Part B: Short Answer – Write legibly in each of the following problems. (30 Pts)

Number of questions: 6 Points per question: 5

B1. Based on the type or types of intermolecular forces, predict **and explain** the substance in each pair that has the higher boiling point:

(a) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$

(a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ has higher b.p. because there are hydrogen bonding between molecules.

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$ has higher b.p. because it has greater dispersion forces.

Part B: Short Answer – Write legibly in each of the following problems. (50 pts)

Number of questions: 6 Points per question: 5

B1. Based on the type or types of intermolecular forces, predict and explain the substance in each pair that has the higher boiling point:

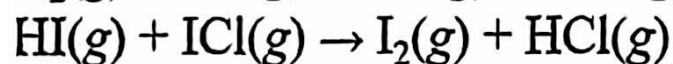
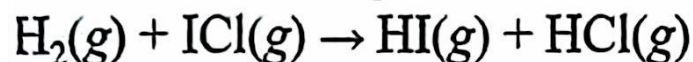
(a) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$

(a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ has higher b.p. because there are hydrogen bonding between molecules.

(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$ has higher b.p. because it has greater dispersion forces.

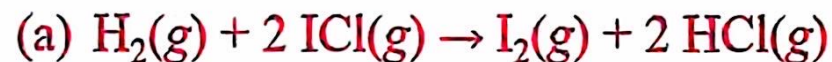
B2. The following mechanism has been proposed for the gas-phase reaction of H_2 with ICl :



(a) Write the balanced equation for the overall reaction.

(b) Identify any intermediates in the mechanism.

(c) If the first step is slow and the second one is fast, which rate law do you expect to be observed for the overall reaction?



(b) HI is the intermediate.

(c) $\text{Rate} = k[\text{H}_2][\text{ICl}]$

B3. List the following aqueous solutions in order of

a) increasing boiling point: 0.12 *m* sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$), 0.05 *m* KCl , 0.05 *m* $\text{Mg}(\text{NO}_3)_2$

b) decreasing freezing point: 0.05 *m* glycerin ($\text{C}_3\text{H}_8\text{O}_3$), 0.04 *m* KBr , 0.04 *m* phenol ($\text{C}_6\text{H}_5\text{OH}$)

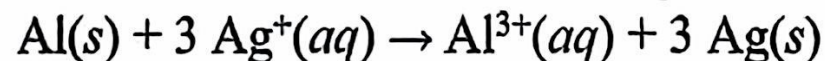
34. How do following changes affect the equilibrium constant K_p for a gas-phase endothermic reaction:

- (a) increase in the total pressure by adding a noble gas
- (b) addition of a reactant
- (c) increase in the temperature

B5. Predict whether the equivalence point of each of the following titrations is below, above, or at pH 7:

- (a) NaHCO_3 titrated with NaOH
- (b) NH_3 titrated with HCl
- (c) KOH titrated with HBr

B6. A voltaic cell utilizes the following reaction:



What is the effect on the cell emf of each of the following changes?

- (a) Water is added to the anode half-cell, diluting the solution.
- (b) The size of the aluminum electrode is increased.
- (c) A solution of AgNO_3 is added to the cathode half-cell, increasing the quantity of Ag^+ but not changing its concentration.

Part C: Long Answer – Write legibly and show work for all steps in the following problems. For mathematical relationships, show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (30 Pts)

Number of questions: 3

Points per question: 10

C1. (a) A 0.10 M solution of formic acid (HCOOH) has pH = 2.38 at 25 °C. Calculate K_a for formic acid at this temperature. (4 Pts)

(b) What is the percentage of the acid that is ionized? (3 Pts)

(c) Predict whether the dipotassium salt of citric acid ($K_2HC_6H_5O_7$) forms an acidic or basic solution in water. For citric acid, $K_{a2} = 1.7 \times 10^{-5}$, $K_{a3} = 4.0 \times 10^{-7}$. (3 Pts)



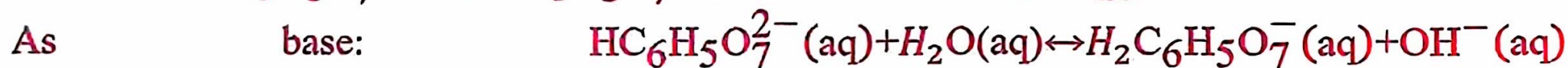
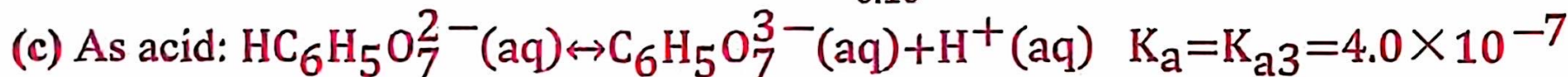
Equilibrium concentration

$$0.10 - x \qquad x \qquad x$$

$$x = [H^+] = 10^{-pH} = 4.2 \times 10^{-3}$$

$$K_a = \frac{[HCOO^-][H^+]}{[HCOOH]} = \frac{x^2}{0.10 - x} = \frac{4.2 \times 10^{-3} \times 4.2 \times 10^{-3}}{0.10 - 4.2 \times 10^{-3}} = 1.8 \times 10^{-4}$$

(b) The percentage ionization of HCOOH = $\frac{x}{0.10} \times 100\% = 4.2\%$



$$K_b = \frac{K_w}{K_{a2}} = \frac{1.0 \times 10^{-14}}{1.7 \times 10^{-5}} = 5.9 \times 10^{-10}$$

The K_{a3} value is almost 10^3 times larger than K_b for $HC_6H_5O_7^{2-}$; thus, the solution is acidic.

C3. The reaction between ethyl iodide ($\text{C}_2\text{H}_5\text{I}$) and hydroxide ion in ethanol ($\text{C}_2\text{H}_5\text{OH}$) solution, has an activation energy of 86.8 kJ/mol and a frequency factor (A) of $2.10 \times 10^{11} \text{ M}^{-1}\text{s}^{-1}$.



(a) Predict the rate constant for the reaction at 40 °C. (3 Pts)

(b) A solution of KOH in ethanol is made up by dissolving 0.343 g KOH in ethanol to form 279.0 mL of solution. Similarly, 1.496 g of $\text{C}_2\text{H}_5\text{I}$ is dissolved in ethanol to form 279.0 mL of solution. Equal volumes of the two solutions are mixed. Assuming the reaction is first order in each reactant, what is the initial rate at 40 °C? (4 Pts)

(c) Assuming the frequency factor and activation energy do not change as a function of temperature, calculate the rate constant for the reaction at 50 °C. (3 Pts)

$$(a) k = Ae^{-E_a/RT} = 2.10 \times 10^{11} \times e^{-86.8 \times 1000 / (8.314 \times 313)} = 6.86 \times 10^{-4} \text{ M}^{-1}\text{s}^{-1}$$

$$(b) [\text{KOH}] = 0.343 \div (56.1 \times 0.279 \times 2) = 0.01096 \text{ M}$$

$$[\text{C}_2\text{H}_5\text{I}] = 1.496 \div (156.0 \times 0.279 \times 2) = 0.01718 \text{ M}$$

$$\text{Rate} = k[\text{KOH}][\text{C}_2\text{H}_5\text{I}] = 6.86 \times 10^{-4} \times 0.01096 \times 0.01718 = 1.29 \times 10^{-7} \text{ M/s}$$

$$(c) k = Ae^{-E_a/RT} = 2.10 \times 10^{11} \times e^{-86.8 \times 1000 / (8.314 \times 323)} = 1.93 \times 10^{-3} \text{ M}^{-1}\text{s}^{-1}$$

$$\text{OR } \ln k_2 = \ln k_1 - \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$= \ln (6.86 \times 10^{-4}) - \frac{86.8 \times 10^3}{8.314} \left(\frac{1}{323} - \frac{1}{313} \right) = 1.033$$

$$k_2 = 1.93 \times 10^{-3} \text{ M}^{-1}\text{s}^{-1}$$

A5. A 0.200 *m* solution of which one of the following solutes will have the lowest vapor pressure?

- A) LiCl B) AlCl₃ C) CaCl₂ D) glucose E) KCl

A6. When solutions of strong electrolytes in water are formed, the ions are surrounded by water molecules. These interactions are described as a case of _____.

- A) hydration B) supersaturation C) crystallization D) dehydration E) saturation

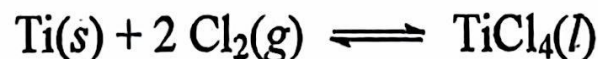
A7. Which one of the following is always positive when a spontaneous process occurs?

- A) $\Delta S_{\text{surroundings}}$ B) ΔS_{system} C) $\Delta S_{\text{universe}}$ D) $\Delta H_{\text{universe}}$ E) ΔG_{system}

A8. Based on electron configuration, which is most likely colorless?

- A) [Ni(NH₃)₆]²⁺ B) [Cd(NH₃)₄]²⁺ C) [Co(NH₃)₆]²⁺ D) [Cu(NH₃)₄]²⁺ E) [Cr(NH₃)₅Cl]²⁺

A9. Consider the following equilibrium.



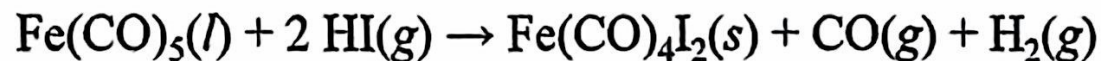
The equilibrium cannot be established when _____ is/are placed in a 1.0 L container.

- A) 0.25 mol Ti(*s*) and 0.25 mol Cl₂(*g*) B) 0.50 mol Cl₂(*g*) and 0.50 mol TiCl₄(*l*)
C) 0.25 mol of Ti(*s*) and 0.25 mol of TiCl₄(*l*) D) 0.75 mol Ti(*s*) E) 1.0 mol TiCl₄(*l*)

A10. With thermodynamics, one cannot determine _____.

- A) the speed of a reaction B) the direction of a spontaneous reaction
C) the extent of a reaction D) the value of the equilibrium constant
E) the temperature at which a reaction will be spontaneous

A11. Which element is oxidized in the reaction below?

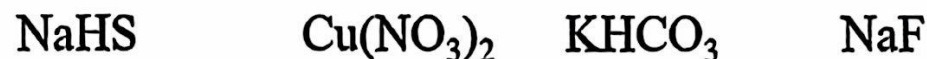


- A) C B) O C) H **D) Fe** E) I

A12. Determine the pH of a 0.35 *M* aqueous solution of CH_3NH_2 (methylamine) at 25.0 ° C. The K_b of methylamine is 4.4×10^{-4} .

- A) 10.00 B) 3.86 **C) 12.09** D) 1.91 E) 13.24

A13. Of the following substances, an aqueous solution of _____ will form basic solutions.

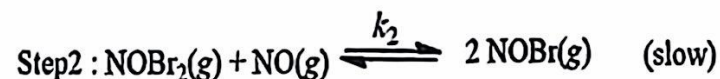


- A) NaHS, $\text{Cu}(\text{NO}_3)_2$ B) KHCO_3 , NaHS C) NaF only D) NaF, KHCO_3 **E) NaHS, KHCO_3 and NaF**

A14. The K_a of benzoic acid is 6.3×10^{-5} . The pH of a buffer prepared by combining 50.0 mL of 1.00 *M* potassium benzoate and 50.0 mL of 1.00 *M* benzoic acid is _____.

- A) 1.71 B) 0.85 C) 3.41 **D) 4.20** E) 2.38

A15. A possible mechanism for the overall reaction: $\text{Br}_2(g) + 2 \text{NO}(g) \rightarrow 2 \text{NOBr}(g)$ is



The rate law for the formation of $\text{NOBr}(g)$ based on this mechanism is rate = _____.

- A) $k_1[\text{NO}]^{1/2}$ B) $k_1[\text{Br}_2]^{1/2}$ C) **$(k_1 k_2 / k_{-1})[\text{NO}]^2[\text{Br}_2]$** D) $(k_1 / k_{-1})^2[\text{NO}]^2$ E) $(k_1 k_2 / k_{-1})[\text{NO}][\text{Br}_2]^2$

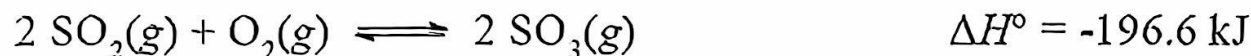
A16. Given the following table of thermodynamic data,

Substance	ΔH_f° (kJ/mol)	S° (J/mol · K)
TiCl ₄ (g)	-763.2	354.9
TiCl ₄ (l)	-804.2	221.9

complete the following sentence. The vaporization of TiCl₄ is _____.

- A) spontaneous at all temperatures
- B) spontaneous at low temperature and nonspontaneous at high temperature
- C) nonspontaneous at low temperature and spontaneous at high temperature
- D) nonspontaneous at all temperatures
- E) not enough information given to draw a conclusion

A17. Consider the following reaction at equilibrium:



Le Châtelier's principle predicts that an increase in temperature will result in _____.

- A) an increase in the partial pressure of O₂
- B) a decrease in the partial pressure of O₂
- C) a decrease in the partial pressure of SO₂
- D) an increase in K_{eq}
- E) no changes in equilibrium partial pressures

A18. In which aqueous system is PbCl₂ least soluble?

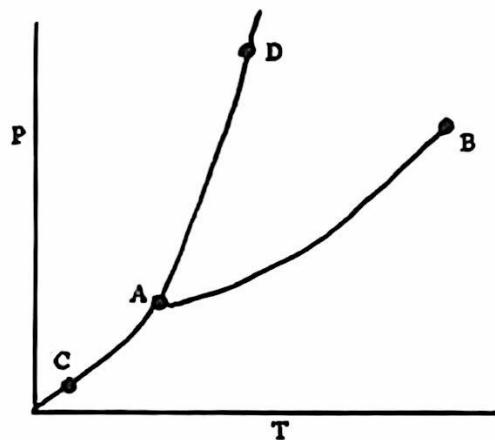
- A) 0.2 M HCl
- B) 0.8 M KCl
- C) H₂O
- D) 0.5 M HCl
- E) 1.0 M HNO₃

Part B: Short Answer – Write legibly and show all work for all steps in the problem. (24 Pts)

Number of questions: 6

Points per question: 4

B1. The phase diagram of a hypothetical substance is



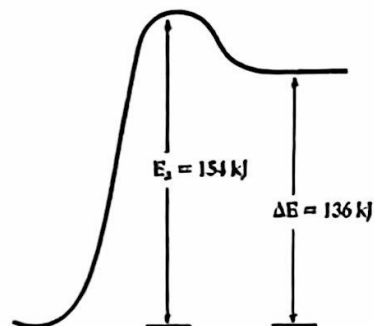
- (a) Which curve (AB/AC/AD) represents a condition of equilibrium between the solid and the gas?
- (b) Which point is the triple point of the substance?

B2. Benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) and aniline ($\text{C}_6\text{H}_5\text{NH}_2$) are both derivatives of benzene. Benzoic acid is an acid with $K_a = 6.3 \times 10^{-5}$ and aniline is a base with $K_b = 4.3 \times 10^{-10}$ at 25.0°C .

- (a) What are the conjugate base of benzoic acid and the conjugate acid of aniline?
- (b) Anilinium chloride ($\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$) is a strong electrolyte that dissociates into anilinium ions ($\text{C}_6\text{H}_5\text{NH}_3^+$) and chloride ions. Which will be more acidic, a 0.10 M solution of benzoic acid or a 0.10 M solution of anilinium chloride?

B3. For the elementary process $\text{N}_2\text{O}_5(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}_3(\text{g})$ the activation energy (E_a) and overall ΔE are 154 kJ and 136 kJ, respectively.

- (a) Draw the Reaction Coordinate Diagrams for this reaction, and label E_a and ΔE .
(b) What is the activation energy for the reverse reaction?



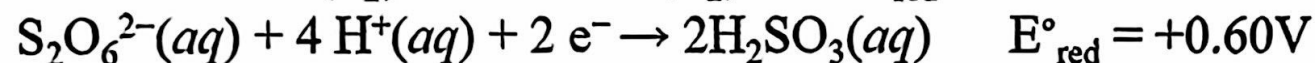
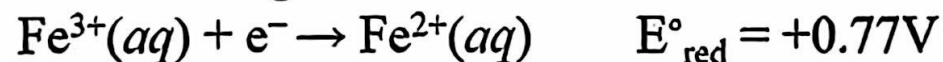
B4. Suppose that the gas-phase reactions $\text{A} \rightarrow \text{B}$ and $\text{B} \rightarrow \text{A}$ are both elementary processes with rate constants of $2.5 \times 10^{-2} \text{ min}^{-1}$ and $2.5 \times 10^{-1} \text{ min}^{-1}$, respectively.

- (a) What is the value of the equilibrium constant for the equilibrium $\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$?
(b) Which is greater at equilibrium, the partial pressure of A or the partial pressure of B?

B5. Predict the stronger acid in each pair: (a) HNO_3 or HNO_2 ; (b) H_2S or H_2O ; (c) H_2SO_4 or H_2SeO_4 ; (d) CH_3COOH or CCl_3COOH .

B6. Write the expression for the solubility product for each of the following ionic compounds: (a) Ag_2CrO_4 ; (b) CaF_2 .

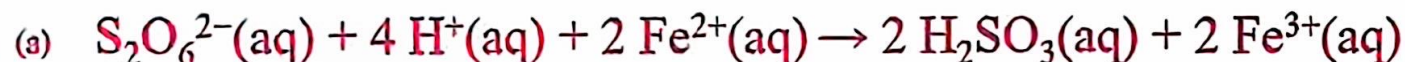
C1. Given the following reduction half-reactions:



(a) Write balanced chemical equation for the oxidation of $\text{Fe}^{2+}(\text{aq})$ by $\text{S}_2\text{O}_6^{2-}(\text{aq})$.

(b) Calculate ΔG° for this reaction at 298 K.

(c) Calculate the equilibrium constant K for this reaction at 298 K.



(b) $E^{\circ} = E^{\circ}_{\text{red}}(\text{cathode}) - E^{\circ}_{\text{red}}(\text{anode}) = 0.60 - 0.77 = -0.17 \text{ V}$

$$\Delta G^{\circ} = -nFE^{\circ} = -2 \times 96485 \times (-0.17) = 33 \text{ kJ/mol OR } 3.3 \times 10^4 \text{ J/mol}$$

(a) $\Delta G^{\circ} = -nFE^{\circ} = -RT \ln K \quad K = e^{\frac{nFE^{\circ}}{RT}} = e^{\frac{2 \times 96485 \times (-0.17)}{8.314 \times 298}} = 1.8 \times 10^{-6} = 10^{-6}$

C2. Cobalt-60 is used in radiation therapy to treat cancer. It has a first-order rate constant for radioactive decay of $k = 1.31 \times 10^{-1} \text{ yr}^{-1}$. Another radioactive isotope, iron-59, which is used as a tracer in the study of iron metabolism, has a rate constant of $k = 1.55 \times 10^{-2} \text{ day}^{-1}$.

(a) What are the half-lives of these two isotopes?

(b) Which one decays at a faster rate?

(c) How much of a 1.00-mg sample of each isotope remains after 3 half-lives?

(d) How much of a 1.00-mg sample of each isotope remains after 5 days?

(a) For a first-order process, $t_{1/2} = 0.693/k$. ^{60}Co : $t_{1/2} = 0.693/1.31 \times 10^{-1} \text{ yr}^{-1} = 5.29 \text{ years}$

^{59}Fe : $t_{1/2} = 0.693/1.55 \times 10^{-2} \text{ day}^{-1} = 44.7 \text{ days}$

(b) Iron-59 decays at a faster rate. (c) For a first-order process, $\ln[A]_t - \ln[A]_0 = -kt$. $\ln[A]_t = -kt + \ln[A]_0$.

$[A]_0 = 1.00 \text{ mg}$; $t = 3 t_{1/2}$. ^{60}Co : $t = 3 t_{1/2} = 3 \times 5.29 = 15.87 \text{ years}$ $\ln [\text{Co}]_t = -1.31 \times 10^{-1} \times 15.87 + \ln (1.00) = -2.08$
 $[\text{Co}]_t = 0.13 \text{ mg}$